

An Interactive Activities of Daily Living Training System with Color-Based Motion Tracking for Early Stage Patients with Alzheimer's Disease

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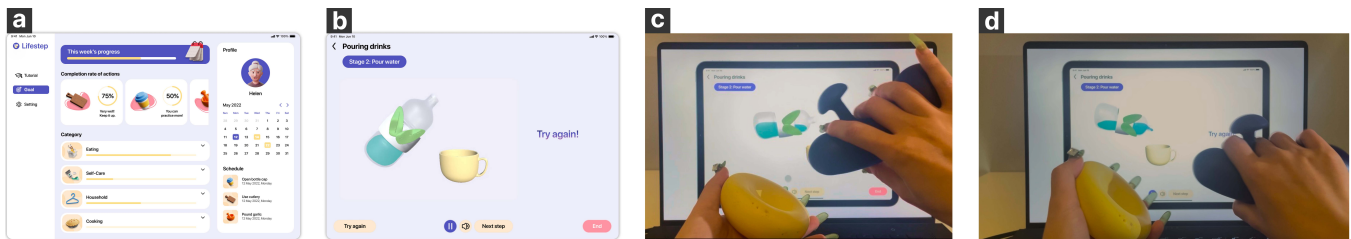


Figure 1: The ADL training demo: (a) illustrates tutorial self-scheduling and progress tracking; (b) shows the practice interface for the pouring water action; (c) and (d) demonstrate how the UI platform supports user to train ADL with the product and provides real-time feedback.

Abstract

Currently, research on Activities of Daily Living (ADL) training for patients with Alzheimer's Disease (AD) remains scarce, with existing approaches focusing primarily on pharmacological treatment or general rehabilitation, lacking comprehensive tools to address cognitive and motor degeneration. In addition, patients' ability to perform ADL is highly dependent on caregivers for urgent assistance, exacerbated caregiver time, and psychological burdens, while not providing sustained improvement for patients. This study introduces an ADL training system tailored for early- to mid-stage AD patients, aimed at mitigating cognitive decline and improving the autonomy and sense of achievement of patients. The system integrates physical interaction products with a user interface platform, employing Python and OpenCV for precise tracking of the motion trajectory to assess movement execution. Furthermore, an animation module developed in Unity provides real-time feedback and dynamic guidance, optimizing the learning experience. By integrating innovative technologies, the system offers a scientifically grounded and systematic approach to training in ADL, thus supporting cognitive maintenance and improving the quality of life for patients with AD.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools; Interactive systems and tools; • Computing methodologies** → *Computer vision.*

Keywords

Alzheimer's Disease; Assistive Technology; Activities of Daily Living (ADL); Motion Tracking

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1 Introduction

Alzheimer's disease (AD) is a neurodegenerative disorder that progressively affects cognition, memory, thinking, and daily functioning. According to the World Health Organization (WHO), approximately 55.2 million people were affected by AD, which disrupts their daily lives and imposes significant physical, financial, and emotional burdens on caregivers [5, 21]. Recent developments in AI-based assistive solutions offer promising avenues to support and recover cognitive functions in patients with AD [11, 19]. However, research gaps remain in real-time judgment, feedback optimization, and interactive experience design for Activity of Daily Living (ADL) training in patients with AD.

This paper presents an interactive training system designed to help people in the early and middle stages of AD participate independently in ADL tasks. The system integrates two physical



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products with a User Interface (UI) platform, employs OpenCV for motion tracking and trajectory analysis, evaluates movement accuracy, and provides real-time feedback. By combining structured guidance with personalized mechanisms, the system provides a low-cost, user-friendly, and interactive solution to improve patient engagement, self-efficacy, and daily independence. The system has the potential to transform AD care by empowering patients, improving their quality of life, reducing caregiver burden, and promoting mental well-being.

The remainder of this paper is organized as follows. Section 2 provides a review of the literature; Section 3 details the design of the system; Section 4 concludes the work and discusses future directions.

2 Literature Review

2.1 Current Status and Challenges of ADL Training for Patients with AD

Studies show that assessing ADL ability is essential for diagnosing AD, monitoring progression, evaluating treatment, and identifying care needs [23]. Training in ADL is gaining attention to support independence, slow cognitive decline, and improve quality of life [18]. In addition, perceived improvement in quality of life influences the acceptance of new technologies by older adults, underscoring the value of tech-based interventions.

Although severe AD patients often rely on professional one-on-one care [26], those in the early to mid stages tend to prefer home care, where familiar settings help memory and task execution [9, 13, 18]. Thus, informal caregivers, typically family members, are crucial [5, 24], but often lack professional guidance, limiting long-term quality of life improvement. Extended care also leads to caregiver stress and health risks [5, 13, 15, 24]. Due to safety risks such as stove misuse or medication errors, constant vigilance is required [24, 25]. In this context, cognitive assistive technologies (CATs) offer potential. For example, the COACH system by Mihailidis et al. [18, 30] uses computer vision and AI for real-time prompting, reducing caregiver intervention and enhancing patient autonomy and engagement.

2.2 Applications of Motion Recognition and Tracking Technology in ADL Training

In ADL training systems, motion recognition and tracking is one of the core modules. Current systems primarily use sensors (e.g. accelerometers, gyroscopes, force sensors) or cameras. Hazman et al. used a wearable glove with resistive and IMU sensors to assess hand motion and reduce development costs [12]. Similarly, Valdivia et al. employed IMU and Kinect V2 for the detection of spinal movements [29]. However, sensor-based systems generally face issues: i) I2C devices may have connectivity problems, with users unable to resolve them; ii) slow progress may cause anxiety, reducing engagement, and data quality. In contrast, camera-based tracking minimizes physical contact and simplifies use. Previous studies have used Python-compatible camera modules such as OpenMV [31], Kinect [6, 16, 29, 32], and Optitrack [16, 32], which offer robust tracking, obstacle avoidance, and data transmission. Color-based tracking

and real-time object detection provide a low-cost but effective way to assess movement accuracy [28].

2.3 Real-time Feedback Mechanism and Multimodal Interaction Design in ADL Training Systems

Real-time feedback is a crucial function in assistive technology. Traditional rule-based systems evaluate actions using preset templates [34], but this approach limits participation, especially among older adults with low acceptance of technology [4]. To improve user experience, newer systems apply real-time data analysis and interaction design, dynamically adjusting guidance based on motion data such as trajectories, joint angles, and speed [1, 22]. For example, Silva et al. developed the Phys.IO app to visualize finger angles using MEMS sensors, enabling sensor-Unity integration [27]. Valdivia et al. showed that different UIs using IMU and Kinect V2 varied in practicality between user groups but were effective for rehabilitation [29]. Multimodal interaction - combining visual cues, animation, and sound - improves task clarity and usability of the system, improving adherence [20]. In summary, integrated multidimensional feedback and interaction design optimize assistive system functionality and experience, providing more comprehensive support to patients with AD.

3 Solution Design

3.1 Selection of Technical Modules

We proposed two technical approaches for user motion capture and evaluation: wearable sensor-based and camera-based color recognition modules. After preliminary testing and comparison, the solution that best aligned with the research objectives was selected.

3.1.1 Wearable Sensor Module. The wearable sensor module integrates the MPU-6050 IMU [1, 12, 14, 29] and ZD-10 force sensors [1] to capture dynamic changes in angle, acceleration and applied force. Using Arduino Uno as the platform, multidimensional motion data is collected and transmitted in real time, supporting motion evaluation and feedback.

The test results show that the high-precision flexible film force sensor reliably outputs force data and fits the human hand, which aids in the optimization of the wearable design. However, the IMU sensor does not communicate due to chip recognition issues, and its high hardware requirements increase cost and complexity.

3.1.2 Camera-based Color Recognition Module. The camera-based solution uses computer vision and OpenCV for real-time color recognition and motion tracking, employing RGB to HSV conversion, color segmentation, morphological operations, and centroid tracking [8, 33] to allow real-time action judgment. To validate the proposed solution, five experimental sets were designed to evaluate its core functions.

Using multiple optimizations (Figure 2), the system can: i) accurately identify and distinguish objects of different colors; ii) achieve stable motion trajectory recognition through data processing, including denoising through morphological operations ('cv2.erode'

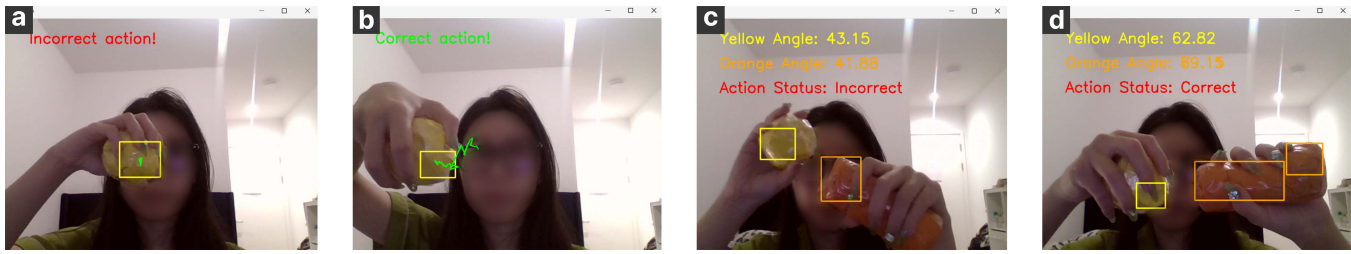


Figure 2: Testing the technical modules: (a) shows the error prompt upon user failure; (b) single object displays successful action feedback, including target motion trajectory and real-time calibration; (c) and (d) show action evaluation of two objects with distinct complex motions, based on dual-color and rotation-angle detection.



Figure 3: Testing the physical products: (a) illustrates simulated ADL motion (e.g., pouring); (b) shows ergonomic testing with prototypes; (c) and (d) present usability testing of motion tracking system by using 3d-printed prototypes.

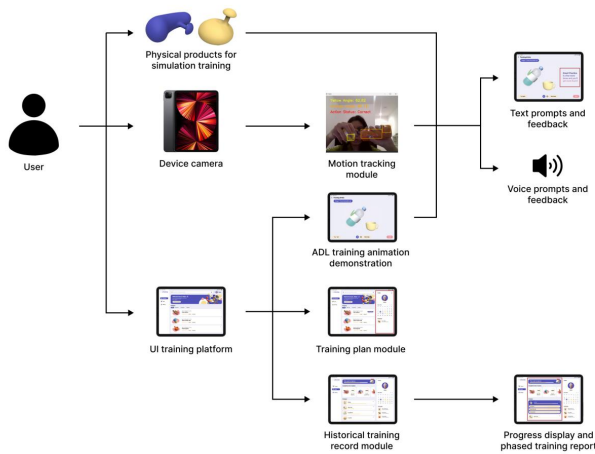


Figure 4: Overall system architecture: the figure illustrates the implementation of multimodal interactive ADL training through the integration of camera-based tracking, physical product interaction, and a user interface platform.

and 'cv2.dilate') and trajectory updates using the 'min' parameter; iii) in compound action judgment, the system tracks motion at fixed points by marking the first time recognized color regions and provides accurate action assessments.

In summary, the system adopts a webcam-based color recognition approach using OpenCV for real-time motion capture and feedback via RGB segmentation, offering a cost-effective, efficient

and user-friendly alternative to complex, unstable IMU and force sensors.

3.2 Physical Product Structure

To support real-time color recognition and motion tracking, the product was ergonomically designed based on common ADL gestures [3] (Figure 3 (a)), with a hemispherical extension to improve the field of view of the camera. The product was refined iteratively through physical prototyping and digital modeling, and printed in 3D (Figure 3 (b)) using flexible resin with a high-contrast blue-yellow color scheme [2, 7, 17] to improve segmentation accuracy. The tests in Python/OpenCV environments confirmed effective color detection and trajectory tracking (Figure 3 (cd)), validating the technical integration of the design.

3.3 UI and Interaction Design

We completed the construction of the UI platform, which allows inclusive access to ADL training through intuitive layouts, adjustable text, and high contrast colors for users [2, 10]. It supports personalized scheduling, real-time motion tracking, and manual or automated feedback (Figure 1 (a)). Developed with Figma and Unity, it integrates animated ADL guidance with audiovisual cues to enhance interaction (Figure 1 (b)). The ongoing work focuses on Unity API–OpenCV integration for adaptive animation and real-time feedback (Figure 1 (c-d)).

3.4 System Structure and Workflow

(Figure 4) summarizes the overall system structure and workflow. The system comprises two main components: the physical product design and the UI platform. Physical design addresses ergonomics

and motion tracking, while user interface design focuses on user interaction and real-time feedback. As a case study for the development of the ADL training system, this work implements the task of "opening a bottle and pouring water" and will try to test more ADLs in the future.

4 Conclusion and Future Work

The proposed ADL training system demonstrates potential in supporting early- to mid-stage patients with AD through real-time color tracking and motion feedback. However, several limitations remain: hardware dependency and individual variability in cognition and mobility affect performance and training consistency; the system's personalization features require further refinement; and the current initial evaluations need long-term validation in real-world settings. Finally, webcam-based image data collection can raise privacy concerns, requiring stronger protective measures.

To address these challenges, future work will improve motion recognition through deep learning, improve hardware adaptability, and expand personalization through behavioral data. The integration of Unity-API and OpenCV will support more dynamic animation feedback, and a caregiver interface will be developed to allow tracking of training progress and assistive planning within home and community care contexts. In addition, comprehensive evaluations will be conducted in real-world environments to assess the system's stability, interaction effectiveness, and training outcomes across diverse user conditions. We aim to refine the system and continue working towards the well-being of patients with AD.

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